Foundations Notebook

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Contents

Chapter 1

Mathematical Foundations

1.1 Calculus

1.1.1 Vector Calculus

Coordinate Syestems

A coordinate system is a method of describing a space that function exists in. The focus will be on that of three dimensional coordinate systems; many examples exist:

1. Cartesian(x,y,z): A point is specified by its coordinates along each basis vector relative to the origin (see Section 1.2.1).
2. Cylindrical(ρ,ϕ,z):
3. Spherical(r,θ, ϕ):

Flux

1.2 Linear Algebra

1.2.1 Vector Spaces

1.2.2 Why This Matters

Linear algebra underpins signal spaces, optimization, and estimation theory.

Chapter 2

Electrical Engineering Foundations

2.1 Electro Magnetism

2.1.1 coulombs Law

2.1.2 Maxwells Equations

$$\nabla \cdot \mathbf{D} = \rho, \tag{2.1}$$

$$\nabla \cdot \mathbf{B} = 0, \tag{2.2}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \tag{2.3}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}, \tag{2.4}$$

where $\mathbf{E}$ is the electric field intensity in the units of $\mathrm{V\,m^{-1}}$, $\mathbf{H}$ is the magnetic field intensity in the units of $\mathrm{A\,m^{-1}}$, $\mathbf{D}$ is the electric flux (see section 1.1.1) densities in the units of $\mathrm{C\,m^{-2}}$, and $\mathbf{B}$ is the magnetic flux density in the units of $\mathrm{Wb\,m^{-2}}$ or a Tesla(T), $\mathbf{D}$ is the electric displacement and (B) is magnetic induction.

2.1.3 Electric Field

This type of electric field is based static charge, and can be thought of like gravity such that the force being exerted is all around if in a 3 d manner that diminishes with the greater range from the source.

$$F = \frac{1}{4\pi\epsilon}\left(\frac{Q_1 Q_2}{r^2}\right) \tag{2.5}$$

2.2 Radar

Radar: Radio Detection And Ranging. Is simply the transmission and receiving of Radio waves. From these radio waves it is possible to determine many things about the incident target.

Simple Model

A transmitter is to be thought as a isotropic radiator with the propagation pattern described by equation 2.6.

$$P_d = \frac{P_t}{4\pi r^2} \tag{2.6}$$

Where P_d is the power density at some radius, P_t is the power transmitted and r is the radius.

$$R = \frac{c\delta t}{2} \tag{2.7}$$

2.2.1 SAR

2.3 Signals and Systems

2.3.1 Adaptive Filtering

What is an Adaptive filter

An adaptive filter uses a Wiener Filter structure but iterates it and adjusts it such that the error is reduced and it begins to approach the ideal filter.

$$h(n+1) = h(n) + \Delta h(n) \tag{2.8}$$

$$\Delta h(n) = -\gamma(\frac{d\sigma^2_{e_{\min}}}{dh}) \tag{2.9}$$

Applications

Using the ides of Wiener Filters and Error Surfaces they can be applied inot the ideas of signal recovery in a noisey channel and noise cancellation.

Noise cancellation Given a the noisy signal and correlated noise we are able to filter the correlated noise such that it matches the noise of the signal and thus allows us to canel it out leaving only the signal in its place.

2.3.2 Sampling

2.3.3 LTI Systems

An LTI system satisfies:

- Linearity
- Time invariance

2.3.4 Idea

- Optical PCB's